

1. Discuss in brief about the differences of below mentioned types of hydropower development schemes.**i. Run-off River scheme****ii. Storage scheme****iii. Pumped storage scheme**

Answer:

Hydropower development schemes are commonly classified as run-of-river, storage, and pumped-storage schemes based on the method of water regulation and utilization.

i. Run-of-river plants:

- RoR hydropower plants do not regulate the hydrograph of the river by one or more seasons.
- The plant utilizes the minimum flow in a river having no appreciable pondage at its upstream.
- In this type of plants, weir or barrage construct across a river, simply to rise and maintain the water level at certain height.
- Usually, low head (up to 70 m) scheme and is suitable only on a perennial river with sufficient dry weather flow to generate designed power.
- Limited storage is used to meet hourly or weekly function of flow.
- No pumping for energy storage.
- Generates power directly from river water.
- Initial cost is generally lower.
- Environmental impact is usually comparatively lower in ROR scheme.
- ROR scheme is applied in base-load and seasonal energy generation.
- Eg: Khimti HPP, Bhotekoshi HPP etc.

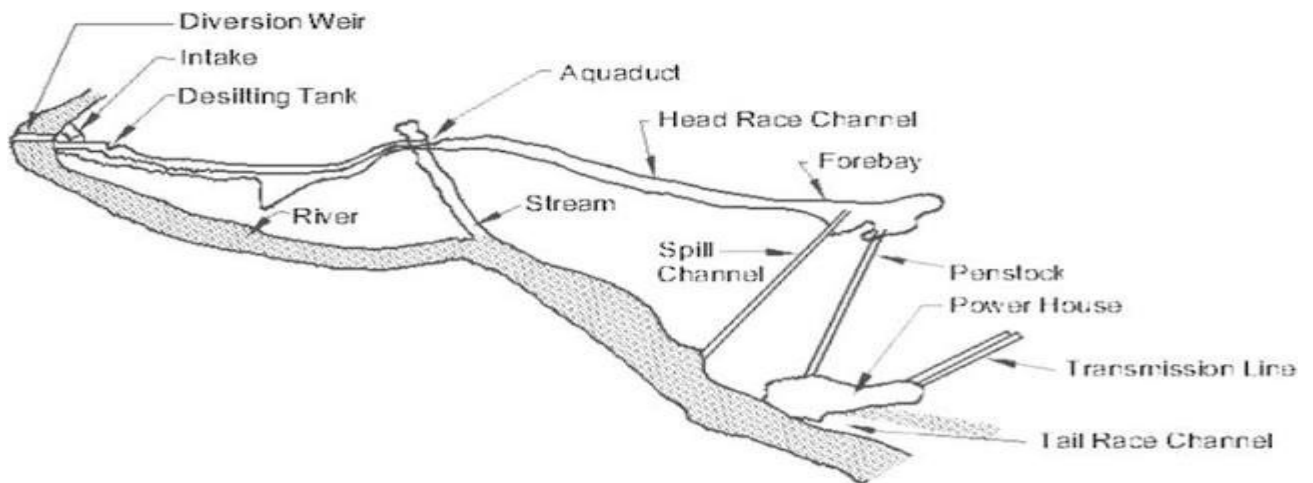
ii. Storage plants:

- Those plants which regulate the hydrograph of the river by one or more seasons are termed as storage plants.
- Powerhouse is located at the foot of dam or away from the dam.
- Water is carried through tunnel to powerhouse.
- It is suitable for medium to high head projects.
- Large reservoir is generally provided.
- More reliable and regulated throughout the year
- Generally, no pumping is required.
- Generates power from stored water.
- Initial cost for storage scheme is high due to dam and reservoir construction.
- Environmental impact due to storage schemes may be significant due to large reservoir and land inundation.
- Storage scheme is applicable for continuous and regulated power generation
- Eg. Kulekhani HPP

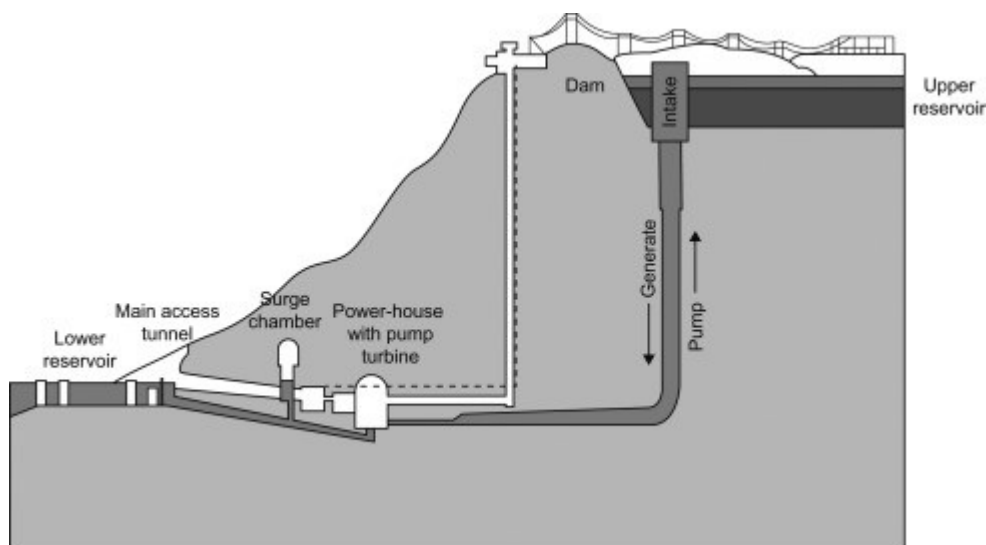
iii. Pump storage plants:

- Plants having water reservoir at upstream as well as downstream of powerhouse, which collects water in upstream reservoir by pumping water from downstream reservoir using cheaper energy.
- Energy is generated during peak hours.
- Requires two reservoirs at different elevations.
- Used mainly for peak-load generation and energy storage.
- Less dependent on natural river flow for short-term operation
- Highly suitable for meeting peak demand
- Consumes electricity for pumping and generates electricity later.

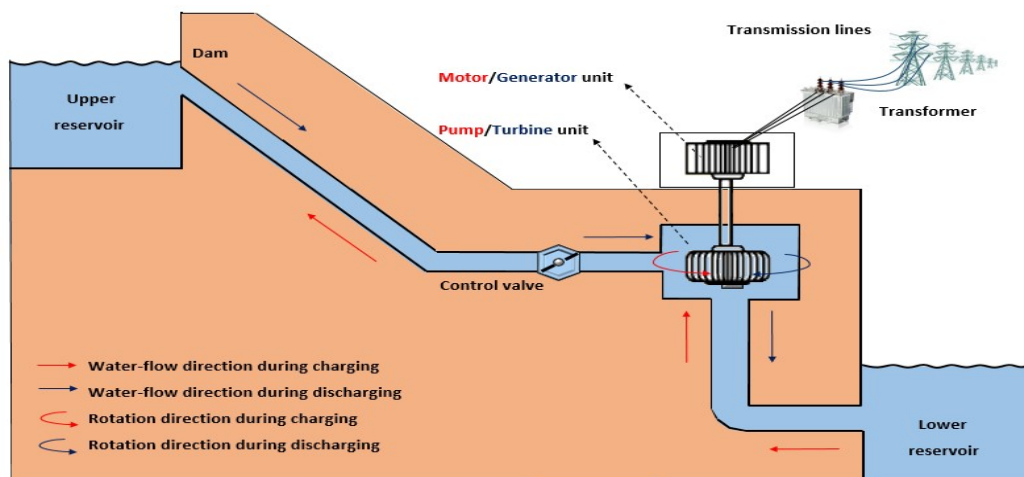
- Initial cost of construction for pumped storage scheme is also high due to reservoirs, pumping equipment, and reversible turbines.
- Environmental impact due to pumped storage scheme depends on the size and location of reservoirs.
- Pumped storage scheme is applicable for peak-load supply, frequency control, and grid balancing.
- Example: Kulekhani-Sisneri Project



a. Run-Off River Scheme



b. Storage hydropower scheme



c. Pumped storage hydropower scheme

2. **Mention some differences between geological and geotechnical investigation works that have to be carried out during field investigation of any hydropower project.**

Answer:

Both geological and geotechnical investigations are essential during the field investigation of a hydropower project. However, geological investigation mainly studies the natural geological conditions of the project area, whereas geotechnical investigation determines the engineering properties of soil and rock for design and construction.

Geological Investigation	Geotechnical Investigation
The main purpose of geological investigation is to understand the geological conditions and history of the project area.	The main purpose of geotechnical investigation is to determine engineering properties of soil and rock.
Main focus of geological investigation are rock types, geological structures, faults, folds, joints, and landslides	Main focus of geotechnical investigation is strength, deformation, permeability, bearing capacity, and other engineering properties
The nature of study is mainly qualitative and regional/site geological assessment.	The nature of study is mainly quantitative engineering assessment.
Geological investigation includes geological mapping, surface observation, identification of faults and discontinuities	Geotechnical investigation includes borehole drilling, test pits, sampling, in-situ and laboratory testing
Rock classification also identifies rock type and geological formation	Geotechnical investigation determines rock mass quality and engineering behavior
Limited engineering testing which focuses mainly on geological interpretation	Extensive laboratory and field tests are conducted in geotechnical investigation.
Selection of project site and identification of geological hazards	Detailed design of foundations and underground structures
It helps select suitable locations for dam, tunnel, powerhouse, and other structures	It helps design dam foundations, tunnel support, slopes, and underground excavations

3. **What do you understand by economic and financial analysis of any runoff river types of hydropower project? Mention its importance.**

Answer:

A run-of-river (RoR) hydropower project generates electricity mainly from the natural flow of a river with little or no large water storage. Economic and financial analyses are carried out to determine whether the project is beneficial to the country and society and financially viable for the project developer and investors.

Economic analysis evaluates the project from the viewpoint of the national economy and society. It compares the overall economic costs and benefits of developing the run-of-river project.

It considers:

- Capital and construction costs
- Operation and maintenance costs
- Value of electricity generated
- Employment and regional development
- Reduction in electricity imports
- Contribution to industrial and economic development
- Environmental and social impacts
- Opportunity cost of resources

The economic viability is commonly evaluated using indicators such as:

- Economic Internal Rate of Return (EIRR)
- Economic Net Present Value (ENPV)
- Economic Benefit-Cost Ratio (B/C Ratio)

Financial analysis evaluates the project from the viewpoint of the developer, investor, and lender. It determines whether the revenue generated from the sale of electricity is sufficient to cover the costs and provide an acceptable return.

It considers:

- Initial investment and construction cost
- Loan and equity financing
- Interest and debt repayment
- Electricity generation and energy sales
- Power purchase agreement (PPA) tariff
- Operation and maintenance costs
- Taxes, depreciation, and insurance
- Seasonal variation in energy generation

Important financial indicators include:

- i. Financial Internal Rate of Return (FIRR)
- ii. Financial Net Present Value (FNPV)
- iii. Benefit-Cost Ratio
- iv. Payback Period
- v. Debt Service Coverage Ratio (DSCR)
- vi. Return on Equity (ROE)

Importance of Economic and Financial Analysis

i. **Determines Project Feasibility**

Helps determine whether the run-of-river project is economically and financially viable.

ii. **Supports Investment Decisions**

Helps investors decide whether to invest in the project.

iii. **Helps in Securing Finance**

Banks and financial institutions use financial indicators to evaluate whether the project can repay its loans.

iv. **Assesses Profitability**

Determines whether expected revenue from electricity generation provides an adequate return.

v. **Evaluates Seasonal Energy Variation**

This is especially important for run-of-river projects because power generation depends directly on river discharge, which varies significantly between the monsoon and dry seasons.

vi. **Helps Select the Optimum Project Size**

Different installed capacities and design discharges can be compared to select the most economical and financially attractive alternative.

vii. **Supports National Planning**

Economic analysis helps the government assess the contribution of the project to energy security, reduced electricity imports, employment, and national development.

viii. **Identifies Project Risks**

The analysis can examine the effects of changes in construction cost, hydrology, electricity tariff, interest rates, and project delays.

4. **What do you understand by economic and financial analysis of any development project? Which one of above-mentioned analysis is generally carried out by the private power producers for the development of hydropower project? Mention some important indicators of financial analysis.**

Answer:

Economic analysis evaluates a project from the viewpoint of the overall national economy and society. It determines whether a project is beneficial to the country by comparing its economic costs and benefits.

It considers:

- Social and economic benefits
- Employment generation
- Contribution to national development
- Environmental and social impacts
- Savings from imports or increase in exports
- Use of national resources
- Economic opportunity cost of resources

Economic analysis generally uses economic prices or shadow prices rather than only market prices.

Financial analysis evaluates a project from the viewpoint of the project developer or investor. It examines whether the project is financially profitable and capable of generating adequate returns.

It considers:

- Initial investment cost
- Construction and operation costs
- Revenue from electricity sales
- Loan and interest payments
- Taxes and depreciation
- Cash flow
- Profitability and return on investment

Financial analysis generally uses actual market prices and financial cash flows.

Private power producers generally carry out financial analysis because their main objective is to determine whether the hydropower project is:

- Financially feasible,
- Profitable,
- Capable of repaying loans, and
- Able to provide an adequate return to investors.

However, economic analysis may also be required by the government or funding agencies for large development projects.

Some important indicators of financial analysis are:

- i. **Net Present Value (NPV)**
The difference between the present value of project benefits and costs.
- ii. **Internal Rate of Return (IRR)**
The discount rate at which the NPV becomes zero.
- iii. **Benefit-Cost Ratio (B/C Ratio)**
Ratio of the present value of benefits to the present value of costs.

- iv. **Payback Period**
The time required to recover the initial investment.
- v. **Debt Service Coverage Ratio (DSCR)**
Indicates the project's ability to generate sufficient cash flow to repay loan obligations.
- vi. **Return on Equity (ROE)**
Measures the return earned by the equity investors.
- vii. **Financial Rate of Return (FRR)**
Measures the financial profitability of the project based on its actual financial costs and benefits.

5. **Describe the importance of reservoir type of project in power system of Nepal.**

Answer:

A reservoir-type hydropower project is a type of project that stores water in a reservoir and releases it in a controlled manner to generate electricity according to demand. This type of project is particularly important for Nepal because most existing hydropower projects are run-of-river (RoR) and their generation decreases significantly during the dry season.

Nepal currently has one major operating reservoir hydropower project: Kulekhani Hydropower Project, which consists of two operating power stations: Kulekhani-I and Kulekhani-II. As of 2026, the most important reservoir-type project actually under construction is Tanahu Hydropower project (140 MW).

Major Importance of reservoir type of project in power system of Nepal are given below:

- i. **Reliable Dry-Season Power Supply**
Reservoir projects can store water during the monsoon and generate electricity during the dry season, reducing seasonal energy shortages.
- ii. **Peak Load Generation**
Water can be released when electricity demand is highest. Therefore, reservoir projects are highly suitable for peak-hour and peak-demand management.
- iii. **Improved Energy Security**
A balanced combination of reservoir and run-of-river projects provides a more reliable and secure national power system.
- iv. **Seasonal Flow Regulation**
Nepal's rivers have high flows during the monsoon and low flows during the dry season. Reservoir storage helps regulate these seasonal variations.
- v. **Support to the National Grid**
Reservoir hydropower can be started and its generation adjusted relatively quickly, helping maintain grid stability, frequency control, and system reliability.
- vi. **Reduction of Energy Import Dependency**
Increased domestic generation during the dry season can reduce dependence on imported electricity.
- vii. **Better Utilization of Run-of-River Projects**
Reservoir projects can complement run-of-river generation by supplying electricity when river discharge and RoR generation are low.
- viii. **Flood Control and Multipurpose Benefits**
In suitable projects, reservoirs can also provide benefits such as flood control, irrigation, water supply, fisheries, and tourism.
- ix. **Facilitates Renewable Energy Integration**
Flexible reservoir hydropower can help balance variable renewable sources such as solar and wind power.

6. **Give the list of electrical and mechanical equipment commonly installed in hydropower plant.**

Answer:

Mechanical equipment converts the energy of water into mechanical energy, while electrical equipment converts this mechanical energy into electrical power and controls, protects, and transmits it to the grid.

Mechanical Equipment:

- i. **Hydraulic Turbine** – Converts water energy into mechanical energy.

- ii. **Penstock** – Conveys water under pressure to the turbine.
- iii. **Main Inlet Valve (MIV)** – Controls and isolates water flow to the turbine.
- iv. **Intake Gates and Stop Logs** – Control and stop water entering the waterway.
- v. **Trash Rack and Trash Cleaning Machine** – Prevent debris from entering the turbine.
- vi. **Surge Tank / Surge Shaft** – Reduces pressure fluctuations and water hammer.
- vii. **Governor System** – Regulates turbine speed and water flow.
- viii. **Draft Tube** – Carries water from the turbine to the tailrace and recovers pressure energy.
- ix. **Cooling Water System** – Provides cooling for various equipment.
- x. **Drainage and Dewatering System** – Removes leakage and accumulated water.
- xi. **Compressed Air System** – Supplies compressed air for control and operation.
- xii. **Lubrication System** – Supplies lubricating oil to turbine and other moving parts.
- xiii. **Cranes and Hoisting Equipment** – Used for installation, maintenance, and handling of heavy equipment.
- xiv. **Fire Protection System** – Detects and controls fire hazards.
- xv. **Ventilation and Air-Conditioning System** – Maintains suitable working conditions in the powerhouse.

Electrical Equipment

- i. **Generator** – Converts mechanical energy from the turbine into electrical energy.
- ii. **Excitation System** – Supplies and controls the generator field current.
- iii. **Generator Transformer (Step-up Transformer)** – Increases generated voltage for transmission.
- iv. **Switchgear** – Controls, protects, and isolates electrical equipment.
- v. **Circuit Breakers** – Interrupt electrical circuits during faults and normal operation.
- vi. **Isolators / Disconnect Switches** – Provide visible isolation for maintenance.
- vii. **Busbars** – Collect and distribute electrical power.
- viii. **Protection Relays** – Detect faults and initiate protective actions.
- ix. **Control and Monitoring System** – Controls and monitors plant operation.
- x. **Instrumentation System** – Measures voltage, current, frequency, pressure, temperature, and other parameters.
- xi. **Station Service Transformer** – Supplies power for auxiliary equipment.
- xii. **Battery and DC Supply System** – Provides reliable power for protection, control, and emergency systems.
- xiii. **Emergency Generator** – Provides backup power during failure of the main supply.
- xiv. **Lightning Protection and Earthing System** – Protects equipment and personnel from lightning and electrical faults.
- xv. **Transmission Lines and Switchyard Equipment** – Transfers generated power to the grid.

7. What are the major benefits of river basin development and integrated water resource development?

Answer:

River basin development means the planned and coordinated development of the water and related resources of an entire river basin for multiple purposes such as hydropower, irrigation, water supply, flood control and navigation.

Integrated Water Resource Development is a comprehensive approach in which surface water, groundwater, land and other related resources are planned and managed together, considering social, economic and environmental needs.

The major benefits are:

- i. **Hydroelectric Power Generation**
Development of rivers for generating clean and renewable electricity.
- ii. **Irrigation Development**
Regulated River flow and reservoirs can provide water for irrigation and increase agricultural production.
- iii. **Flood Control**
Reservoirs and other storage structures can help regulate flood flows and reduce flood damage downstream.
- iv. **Drinking and Industrial Water Supply**
Stored and regulated water can be supplied for domestic, municipal, and industrial purposes.
- v. **Navigation**
In suitable rivers, flow regulation and water storage can improve navigation facilities.

- vi. **Fisheries Development**
Reservoirs may support fisheries and provide additional economic opportunities for local communities.
- vii. **Recreation and Tourism**
Reservoirs and surrounding areas can promote boating, fishing, tourism, and recreational activities.
- viii. **Employment and Regional Development**
Construction and operation of hydropower and other basin projects create employment and promote infrastructure development.
- ix. **Soil and Watershed Conservation**
Integrated basin development may include afforestation, erosion control, and watershed management, reducing sedimentation.
- x. **Improved Water Resource Management**
River basin development enables coordinated and optimum utilization of available water resources for multiple sectors.

8. Describe concept of IEE and EIA.

Answer:

Initial Environment examination (IEE):

- According to Environment Protection Act (EPA) 2076 and Environment Protection Rules (EPR) 2077, IEE is defined as "An analytical study or evaluation conducted to determine whether a proposed project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts."
- Done for medium scale projects that has moderate effect on environment.
- Project requiring IEE are listed in Schedule 2 of EPR 2077.
- Includes:
 - Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → IEE Report Preparation → Recommendation and submission → final review and approval → Decision making
- If IEE Report submitted is approved by concerned authority, the project is implemented.
- IEE is done for:
 - a. Highways with the length of road 5-50 km
 - b. Irrigation project with command area 25-2000 hectares.
 - c. Hydropower project with installed capacity of 1 MW-50 MW

Environmental Impact assessment (EIA):

- "Environmental impact assessment" is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
- Projects requiring EIA are listed in schedule 3 of EPR 2077.
- Includes:
 - Screening → Scoping → Preparation and approval of work schedule → Field study and public hearing → EIA Report Preparation → Recommendation and submission → final review and approval → Decision making
- EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).
- EIA is done for:
 - a. Highways with the length of road greater than 50 km
 - b. Irrigation project with command area greater than 2000 hectares.
 - c. Hydropower projects with installed capacity of greater than 50 MW.

9. What do you mean by screening and ranking of hydropower projects? Discuss its advantages and disadvantages.

Answer:

Screening of hydropower projects is the preliminary process of examining a large number of potential projects and eliminating those that are technically, economically, environmentally, or socially unsuitable. It helps identify feasible projects for further detailed investigation.

Ranking of hydropower projects is the process of comparing the screened feasible projects using selected criteria and arranging them in order of priority. The criteria may include:

- Installed capacity and energy generation
- Construction cost
- Economic benefits and financial viability
- Hydrological potential
- Environmental and social impacts
- Accessibility and infrastructure requirements
- Construction difficulty and geological conditions
- Contribution to national energy demand
- Project implementation time

Thus, screening identifies suitable projects, whereas ranking determines their priority for development.

Advantages:

- i. **Efficient Project Selection**
Helps identify the most suitable hydropower projects from a large number of alternatives.
- ii. **Better Allocation of Resources**
Limited financial, technical, and human resources can be directed towards high-priority projects.
- iii. **Saves Time and Cost**
Unsuitable projects can be rejected during the preliminary stage, avoiding unnecessary detailed investigations.
- iv. **Supports Decision-Making**
Provides a systematic basis for government agencies and investors to prioritize projects.
- v. **Considers Multiple Factors**
Technical, economic, environmental, and social aspects can be considered together.
- vi. **Promotes Planned Development**
Helps develop hydropower resources in a systematic manner according to national priorities.
- vii. **Reduces Investment Risk**
Projects with major constraints can be identified early.

Disadvantages:

- i. **Dependence on Available Data**
Preliminary screening and ranking may be inaccurate when sufficient hydrological, geological, or environmental data are unavailable.
- ii. **Subjectivity in Criteria and Weighting**
The selection and weighting of criteria may influence the final ranking.
- iii. **Changes Over Time**
Project ranking may change due to variations in electricity demand, construction costs, technology, environmental regulations, or market conditions.
- iv. **May Oversimplify Complex Projects**
Complex technical, environmental, and social issues may not be fully represented by numerical ranking.
- v. **Uncertainty in Preliminary Estimates**
Early estimates of project cost, energy generation, and benefits may differ significantly from actual values.

10. What is optimization? Describe the principles of optimization of a water conveyance system.

Answer:

Optimization is the process of obtaining the best possible solution from a number of available alternatives by achieving the desired objective while satisfying specified constraints. In a water conveyance system,

optimization aims to determine the most economical and efficient combination of system components—such as pipe diameter, material, alignment, pumping arrangement, and operating conditions—to deliver the required quantity of water safely and reliably.

Principles of Optimization of a Water Conveyance System

The main principles are:

- i. **Minimum Total Cost**
The system should be designed to minimize the total cost, including:
 - a. Initial construction cost,
 - b. Pipe and equipment cost,
 - c. Pumping and energy cost,
 - d. Operation and maintenance cost.
- ii. **Adequate Hydraulic Capacity**
The conveyance system should carry the required design discharge without excessive head loss or inadequate pressure.
- iii. **Economical Pipe Diameter**
The optimum pipe diameter should balance:
 - a. Higher initial cost of larger pipes, and
 - b. Higher friction and pumping costs of smaller pipes.
- iv. **Minimum Energy Requirement**
The system should minimize unnecessary head losses and pumping energy through efficient pipe sizing, alignment, and equipment selection.
- v. **Proper Alignment and Route Selection**
The conveyance route should be as short and economical as possible while avoiding unstable terrain, environmentally sensitive areas, and unnecessary elevation changes.
- vi. **Reliability and Safety**
The system should ensure continuous water supply and be designed to withstand expected pressure, surge, leakage, and other operational conditions.
- vii. **Efficient Use of Resources**
Materials, energy, land, and water resources should be used efficiently with minimum wastage.
- viii. **Minimum Environmental Impact**
The selected system should reduce adverse effects on the environment and surrounding communities.
- ix. **Ease of Operation and Maintenance**
The design should allow easy inspection, repair, maintenance, and future expansion.

11. In view of existing safety rules and regulation, what are the safety measures which need to be adopted in storage and handling of explosive in construction of a hydropower project?

Answer:

In hydropower projects, explosives may be required for tunnel excavation, road construction, quarrying, and other rock excavation works. Explosives used in hydropower and major infrastructure projects are strictly controlled. The Nepal Army has an important role in the security, controlled blasting and disposal of explosives, and hydropower projects coordinate with trained/authorized Army personnel for such operations. Proper storage, transportation, handling, blasting and disposal procedures are essential because explosives can cause fatal accidents.

Recent examples: On 10 August 2026, an explosion occurred at an Army post in Rasuwa during the disposal of expired explosives associated with hydropower/road blasting. One person was killed and six were injured, including two Army personnel. Another recent example is the Upper Trishuli 3A project in Nuwakot, where Nepal Army engineers carried out controlled blasting during rescue operations to clear blocked tunnel sections after the August 2026 floods.

Considering applicable safety rules and regulations, the following safety measures should be adopted:

- i. **Licensed and Authorized Personnel:**
Explosives should be handled only by trained, competent, and authorized personnel.
- ii. **Safe Storage Magazine:**
Explosives should be stored in a properly designed, approved, and secured explosive magazine located away from residential areas, offices, fuel storage, and other hazardous locations.
- iii. **Separate Storage:**
Detonators, detonating cords, and other initiating devices should be stored separately from explosives to prevent accidental initiation.
- iv. **Access Control and Security:**
The explosive storage area should be securely fenced, locked, and guarded. Entry should be restricted to authorized persons only.
- v. **Fire and Ignition Prevention:**
Smoking, open flames, welding, and other sources of ignition should be strictly prohibited near explosive storage and handling areas.
- vi. **Proper Transportation:**
Explosives should be transported in approved vehicles and containers. Explosives and detonators should not be transported together unless permitted under applicable regulations.
- vii. **Careful Handling:**
Explosives should be protected from shock, friction, excessive heat, moisture, and mechanical damage. Packages should not be dropped, thrown, or roughly handled.
- viii. **Warning Signs and Communication:**
Appropriate danger signs, warning systems, and restricted-area notices should be provided. Effective communication should be maintained during blasting operations.
- ix. **Blasting Safety Procedures:**
Before blasting, the area should be cleared and guarded, warning signals should be given, and all workers should move to a safe location.
- x. **Misfire Management:**
Misfires should be handled only by qualified blasting personnel according to approved procedures. No unauthorized person should enter the blast area until it is declared safe.
- xi. **Record Keeping and Inventory Control:**
Proper records of the receipt, storage, issue, use, and balance of explosives should be maintained.
- xii. **Emergency Preparedness:**
A fire protection and emergency response plan should be prepared. Workers should be trained in emergency procedures and reporting accidents.